

Q1. Let N_1 and N_2 be 3×3 nilpotent matrices over $\mathbb{F}$.
Show that N_1 and N_2 are similar if and only if they have same minimal polynomial.

Q2. If A is a complex matrix with characteristic polynomial $(x-2)^3(x+7)^2$ and minimal polynomial $(x-2)^2(x+7)$, what is the Jordan form of A ?

Q3. How many possible Jordan form are there for a complex matrix with characteristic polynomial $(x+2)^3(x-1)^2$?

Q4. Classify up to similarity of all 3×3 complex matrices A such that $A^3 = I$.

Q5. Let N_1 and N_2 be 6×6 nilpotent matrices over $\mathbb{F}$. Suppose that N_1 and N_2 have the same minimal polynomial and the same nullity ($\dim(N(N_1)) = \dim(N(N_2))$). Prove that N_1 and N_2 are similar.

Q6. Let $J(m)$ be the Jordan matrix of size m . Show that $J(m)$ and $J(m)^T$ is similar. Use Jordan canonical form to prove that every complex matrix A is similar to A^T .